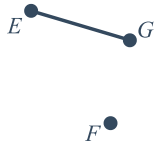


## Line segments and distance



1



2  $\overline{AC}$

3  $\overline{XW}$

4  $\overline{AE}$

5  $\overline{DF}$

6

a 5 units

b 5 units

7

a 4 units

b 1 unit

c  $c = \sqrt{17}$

8

a 2 units

b 1 unit

9  $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

10 Distance =  $\sqrt{(-5 - 4)^2 + (10 - (-5))^2}$

11 10.00

12

a 15 units

b  $\sqrt{73}$  units

13  $\sqrt{34}$  units



14

- a  $\sqrt{82}$  units
- b  $\sqrt{26}$  units
- c  $N$

15

- a  $\sqrt{50}$  units
- c  $M$

b 5 units

16

- a  $\sqrt{50}$  units
- b  $\sqrt{2}$  units
- c 5 : 1

17

a  $CE = 28$

b  $GH = 15$

18

a  $QS = 63$

b  $PQ = 24$

19

a  $x = 3$

b 25

20  $JT = 40$

21

- a
  - i  $x = 3$
  - ii 6
- b
  - i  $x = 3$
  - ii 3



## Additional questions

22

a

- i  $AB = 3$  units
- ii  $BC = 4$  units
- iii  $AC = 5$  units

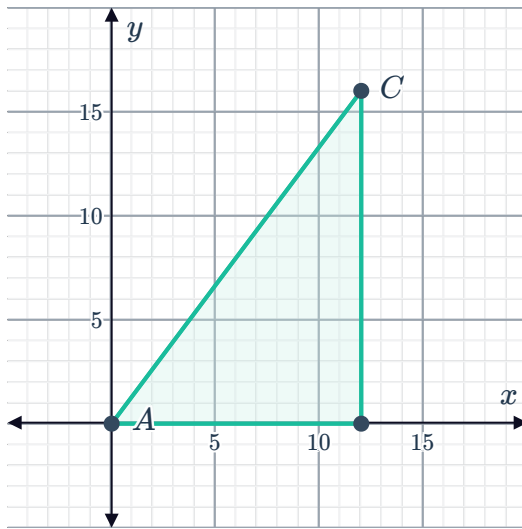
b

- i  $AB = 4$  units
- ii  $BC = 7$  units
- iii  $AC = 8.06$  units

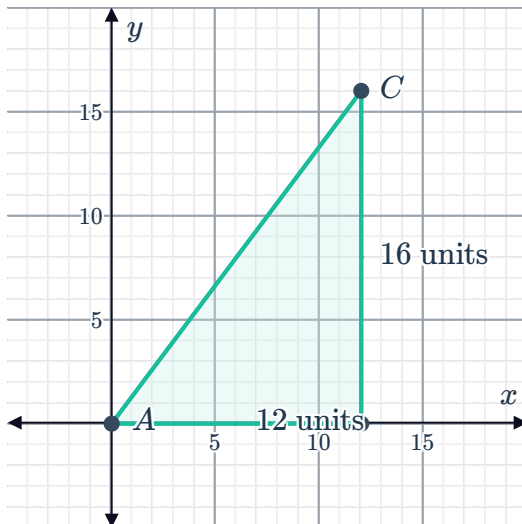
23

a

i



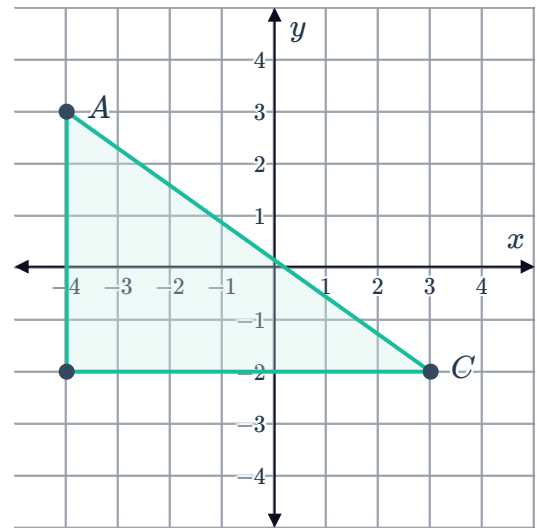
ii



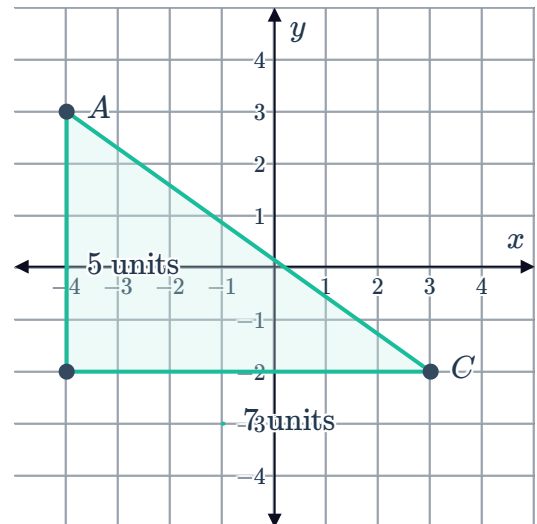
- iii  $AC = 20$  units

b

i



ii



- iii  $AC = 8.6$  units

24



25

- a 10 units
- b 15 units
- c 8.54 units
- d 5.83 units

26

- a  $PM = 9.90$  units
- b  $PN = 11.18$  units
- c  $N$

27 17 units

28

- a  $AC = 14.14$  units
- b  $BD = 8.49$  units

29

- a Culmer Station is the closest to their meeting spot.
- b When actually walking, we need to consider traffic lights, side walks and what is accessible. Depending on which direction they are catching the train may also impact their choice.
- c If they need to stop at the Winn-Dixie, then the Civic Center Station would have the shortest trip distance.

30

- a There are two possible coordinates.
- b  $M(-2, 9)$  and  $M(-2, 3)$
- c If we were not told the line segment was vertical, there would be an infinite number of answers.

31 The distance formula comes from the Pythagorean theorem.  $(x_2 - x_1)$  is the calculation for the length of the line segment which is the horizontal component of the right triangle we can create.  $(y_2 - y_1)$  is the calculation for the length of the line segment which is the vertical component of the right triangle we can create.

32 Both Dayanna and David are correct.  $(x_2 - x_1)$  and  $(x_1 - x_2)$  will have the same magnitude, but different signs. However, when we square them, they will both become positive, so they will give the same answer.

33



- a Fyfe made an error on Step 2. They incorrectly substituted  $(x_2 - y_2)$  and  $(y_1 - x_1)$ , but this calculation does not connect to the Pythagorean theorem to make a right triangle and so does not find the distance.

b

$$\begin{array}{ll}
 \textcircled{1} & AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad \text{Distance formula} \\
 \textcircled{2} & = \sqrt{(3 - -4)^2 + (-5 - 7)^2} \quad \text{Substitution Property of Equality} \\
 \textcircled{3} & = \sqrt{(7)^2 + (-12)^2} \quad \text{Simplify} \\
 \textcircled{4} & = \sqrt{49 + 144} \quad \text{Simplify} \\
 \textcircled{5} & = \sqrt{193} \quad \text{Simplify}
 \end{array}$$

34

- |   |          |   |          |
|---|----------|---|----------|
| a | $AC = 1$ | b | $AE = 4$ |
| c | $EF = 1$ | d | $FB = 2$ |
| e | $DB = 4$ | f | $AB = 7$ |

35

- |   |                                                                                |   |                                                                                |
|---|--------------------------------------------------------------------------------|---|--------------------------------------------------------------------------------|
| a | By the segment addition postulate we know that $CD + DE = CE$ and so $CE = 28$ | b | By the segment addition postulate we know that $FG + GH = FH$ and so $GH = 15$ |
|---|--------------------------------------------------------------------------------|---|--------------------------------------------------------------------------------|

36

- |   |                       |   |                        |
|---|-----------------------|---|------------------------|
| a | $PR = 5.1 \text{ mm}$ | b | $AB = 51.4 \text{ cm}$ |
|---|-----------------------|---|------------------------|

37

- a  $\overline{DE}$  and  $\overline{EF}$
- b  $\overline{CF}$  and  $\overline{DB}$
- c In part (a) you would find two new congruent segments,  $\overline{CG}$  and  $\overline{DG}$ , and in part (b) you would not find any additional congruent segments.

38

a  $KL = 25$

b  $JT = 40$



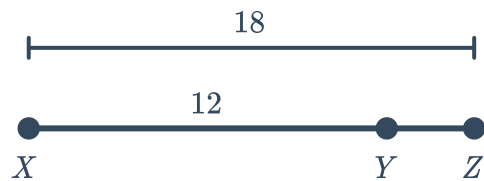
39

a  $BD = 12$

b  $CB = 3$

40  $ML = 51$

41  $YZ$  could be 30 units in length or 6 units in length:



42 By the segment addition postulate, if points  $A$ ,  $B$ , and  $C$  were collinear, then  $AB + BC = AC$ .

Since  $AB + BC = 15$  and  $AC = 12$ , we know that these points cannot be collinear.